Name: _____

Investigating refraction

core practical

Date:

Time:

Total marks available: 25

Total marks achieved: _____

Questions

Q1.

A student investigates what happens when light travels from air to glass.

Figure 15 shows some of the apparatus used in the investigation.

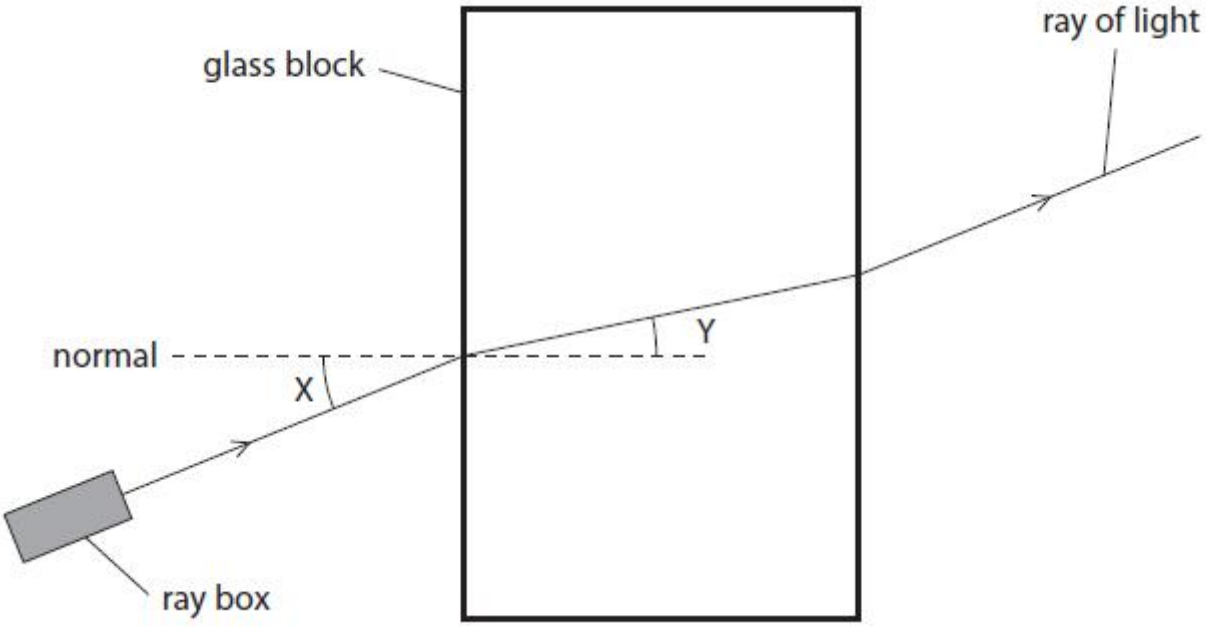

Figure 15

(i) In Figure 15, angle Y is the angle of

(1)

- ☒ **A** deflection
- ☒ **B** incidence
- ☒ **C** reflection
- ☒ **D** refraction

(ii) Figure 16 is a graph of the student's results.

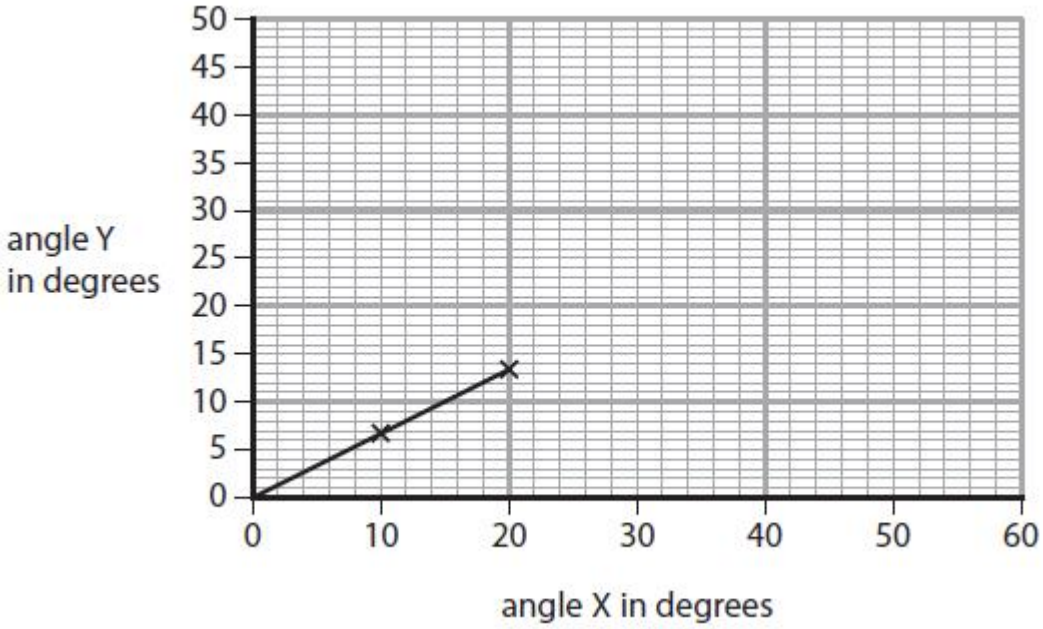

Figure 16

Use the graph to calculate a value for

$$\frac{\text{angle Y}}{\text{angle X}}$$

(2)

$\frac{\text{angle Y}}{\text{angle X}} =$ .....

(iii) The student concludes that angle Y is directly proportional to angle X.

Explain what the student must do to test this conclusion in more detail.

(3)

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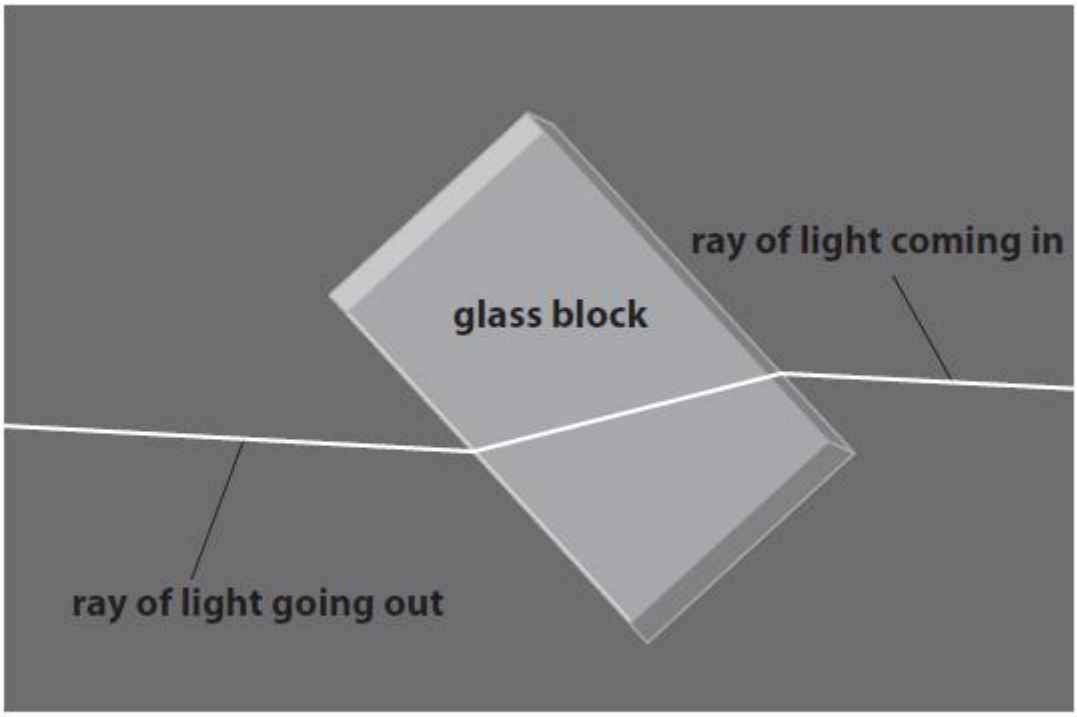
(Total for question = 6 marks)

Q2.

Figure 2 is a diagram of a glass block being used to investigate the refraction of light.



(Source: © ajizai/docstoc.com)

Figure 2

The angle of incidence was varied and the angle of refraction was measured each time.

Figure 3 shows the results in a table.

angle of incidence in degrees	angle of refraction in degrees
20	13
30	20
40	25
50	31
60	35
70	39
80	41

Figure 3

Figure 4 shows a graph of these results.

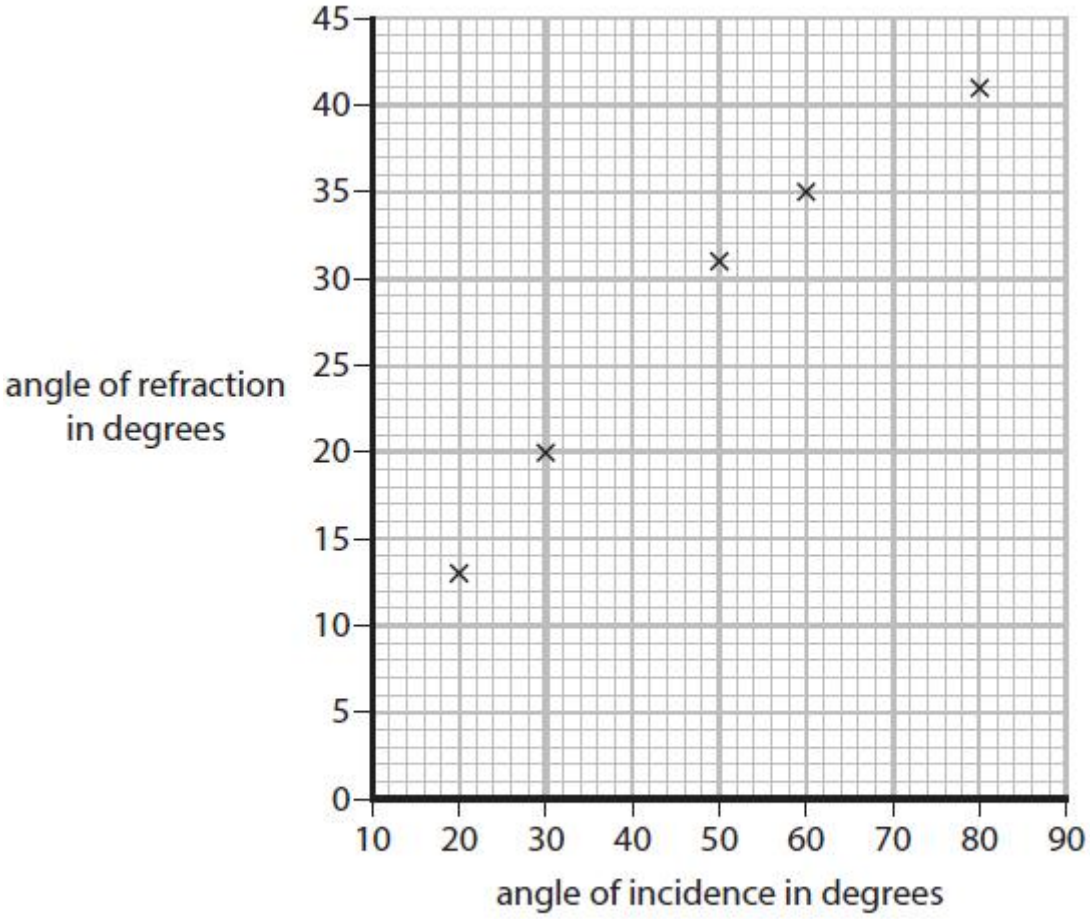


Figure 4

(i) The two shaded rows in Figure 3 are not plotted on the graph in Figure 4.
Plot these missing results on the graph in Figure 4.

(2)

(ii) Draw a curve of best fit through the points on the graph in Figure 4.

(1)

(i) Using Figure 4, describe the relationship between the angle of refraction and the angle of incidence.

(2)

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(Total for question = 5 marks)

Q3.

A student investigates how the angle of refraction varies with the angle of incidence when light enters a glass block.

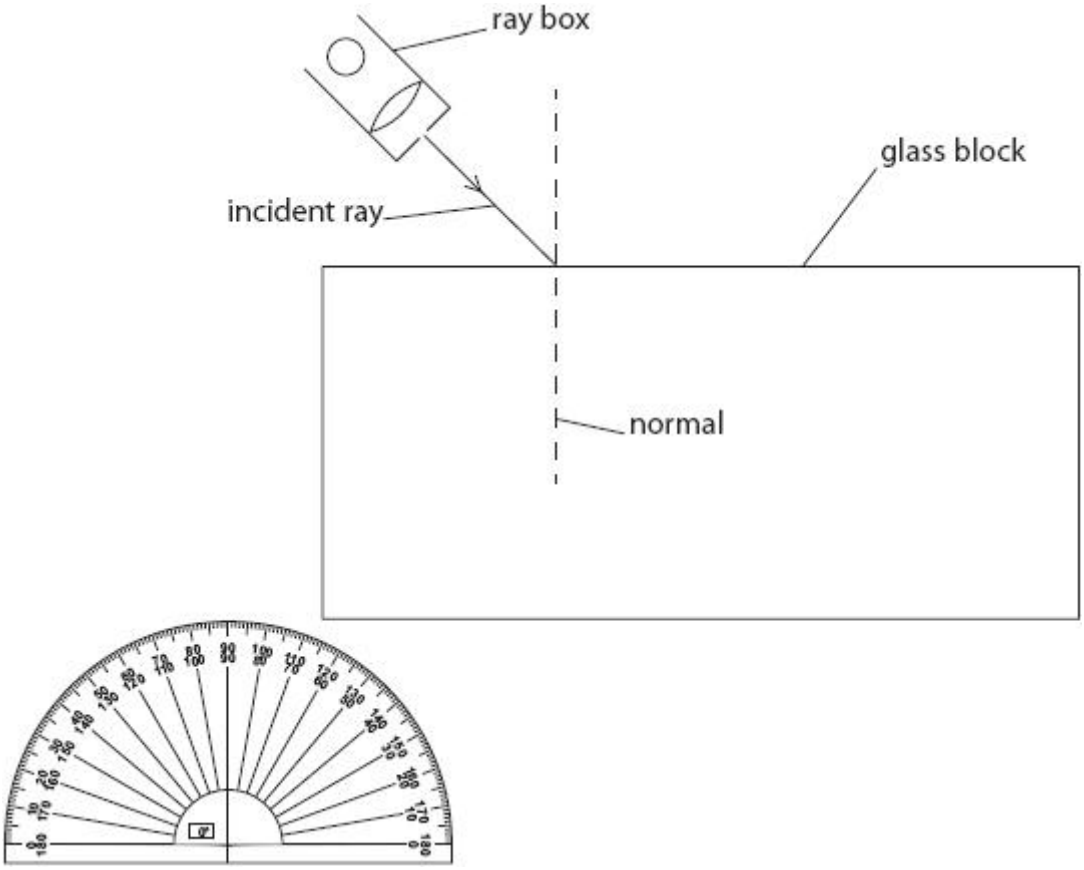

Figure 5

Describe how the student should perform this investigation.

(4)

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(Total for question = 4 marks)

Q4.

A student investigates how the angle of refraction varies with the angle of incidence when light enters a glass block.

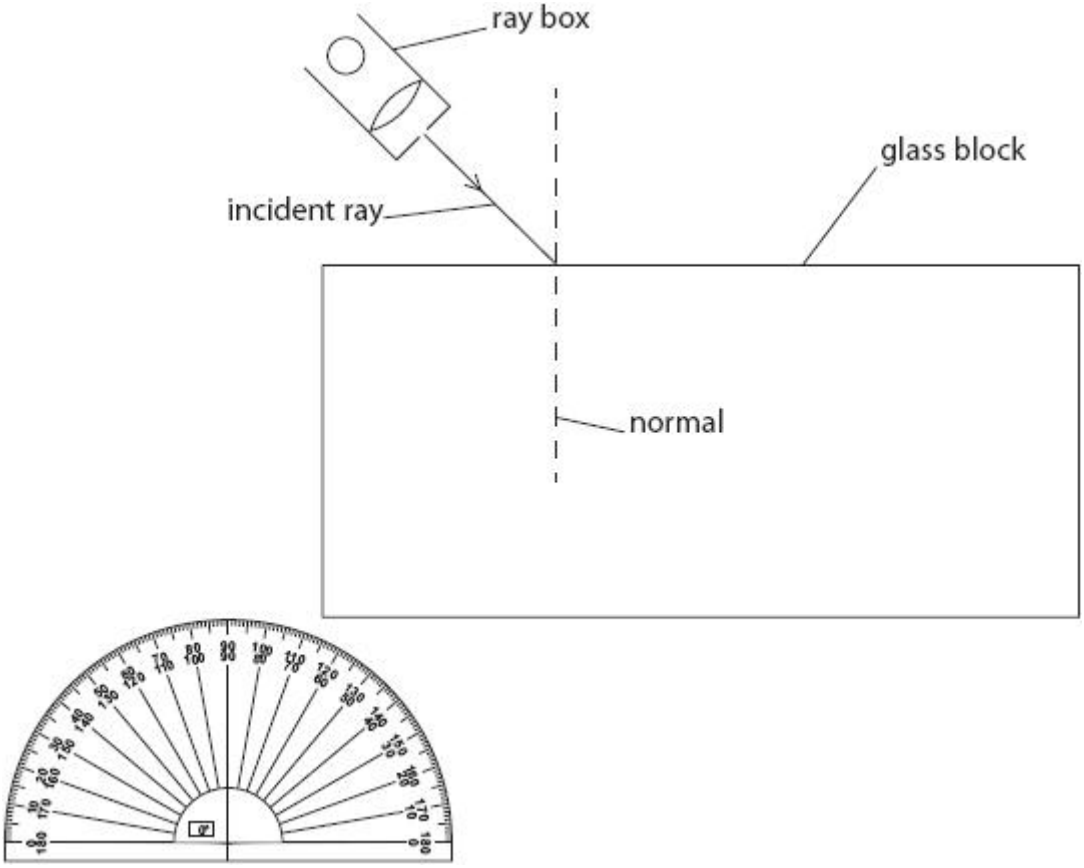

Figure 5

The student's results are shown in Figure 6.

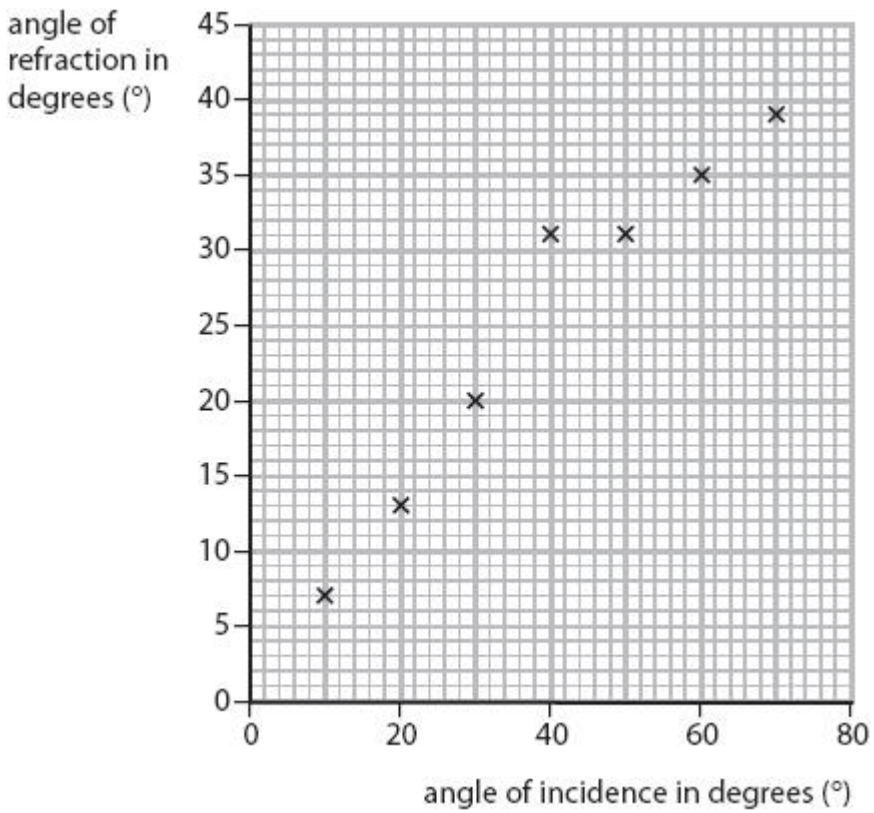

Figure 6

Explain whether any of the readings are anomalous.

(2)

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(Total for question = 2 marks)

Q5.

When light travels from air into glass, it is refracted.

To investigate this, a student has a rectangular glass block and a ray box.

The ray box produces a ray of light.

Describe how the student should use the glass block and ray box to produce a refracted ray and mark its position.

You may draw a diagram to help your answer.

(4)

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(Total for question = 4 marks)

Q6.

A student is investigating the refraction of light.

Figure 4 shows part of the apparatus and the angles to be measured.

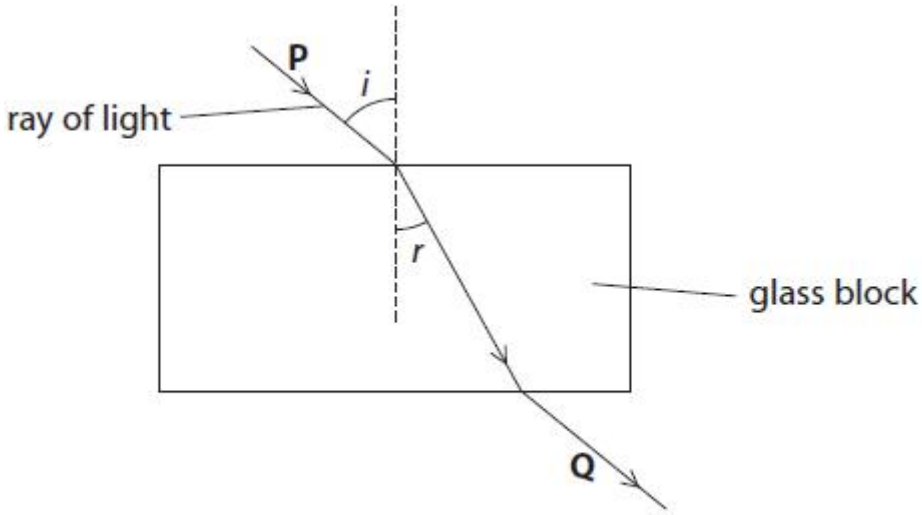

Figure 4

The student measures angle *r* for several different values of angle *i*.

Figure 5 shows the student's results.

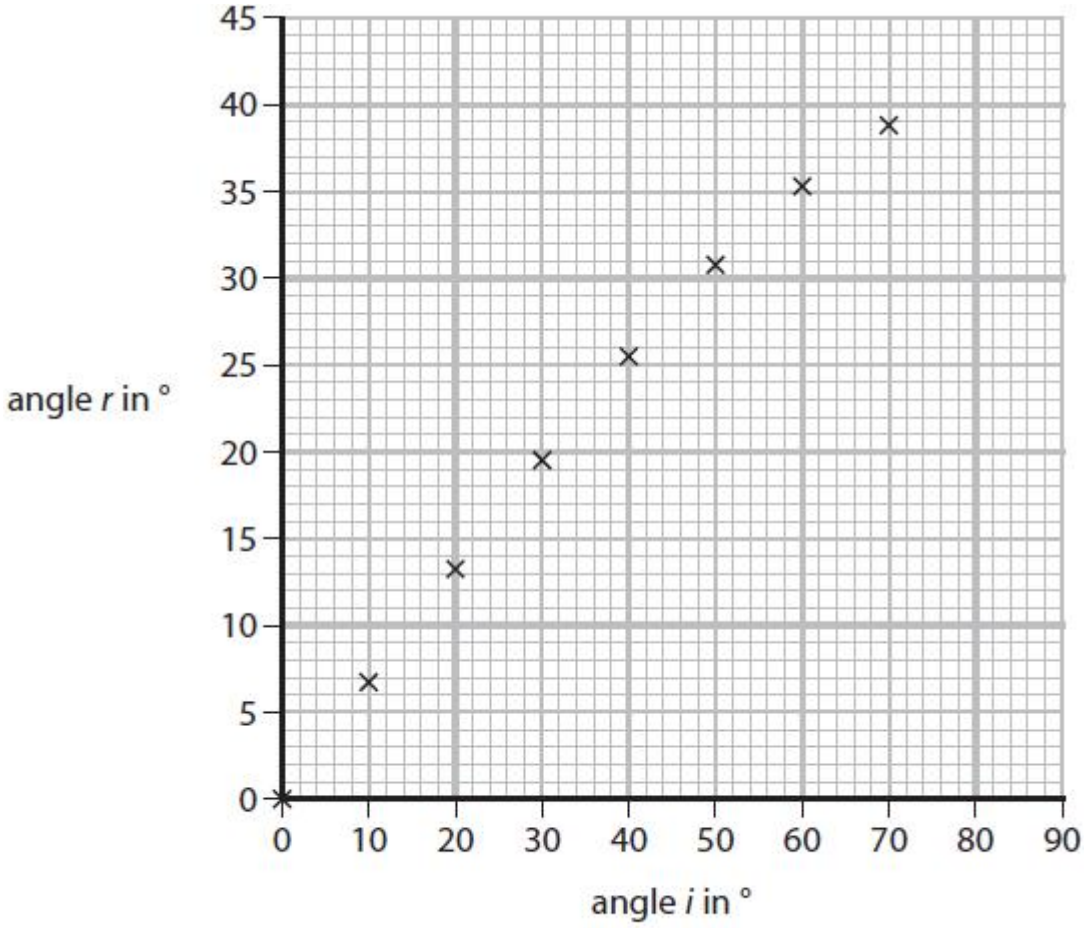

Figure 5

(i) On the graph in Figure 5, draw the best fit curve.

(1)

(ii) Use the graph in Figure 5 to estimate the value of angle *r* when angle *i* is 80°.

(1)

angle *r* = ... °

(iii) Describe how angle *r* changes with angle *i* for the results shown on the graph in Figure 5.

(2)

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(Total for question = 4 marks)

Mark Scheme

Q1.

Question Number	Answer	Mark
(i)	**D** refraction **The only correct answer is D** *A 'deflection' is an incorrect distracting description* *B 'incidence' is incorrect, that would be angle X* *C 'reflection' is incorrect, no reflection being shown in the diagram*	**(1)** AO 1 1

Question Number	Answer	Additional guidance	Mark
(ii)	any pair of coordinates selected from the line (1) in range → 0.6(0) to 0.7(0) (1)	e.g. 20 and (13 or 14) or 10 and (6 or 7) ignore any units given award full marks for a correct answer without working	**(2)** AO 2 1

Question Number	Answer	Additional guidance	Mark
(iii)	an explanation linking: repeat (1) different angles / more values of X (1) for larger angles / values of X (1)	 allow 'more measurements' / 'repeat experiment' / collect more data > 20°	(3) AO 3 3a

Q2.

Q3.

Question number	Answer	Additional guidance	Marks
	An answer that combines the following to provide a logical description of the method	Responses may be seen on diagram. allow 'calculate' for 'measure' in context ('measure angles' insufficient – must be specified)	**(4)**
	• measure angle of incidence (1)	measure angle from incident ray	
	• measure angle of refraction (1)	measure angle from refracted ray accept (reference to drawing path of) refracted ray (inside block) / emergent ray / ray under or leaving block	
	• measure angle(s) to the **normal** (1)	allow to 'dotted line' for normal	
	• repeat for different angle(s) (of incidence) (1)		

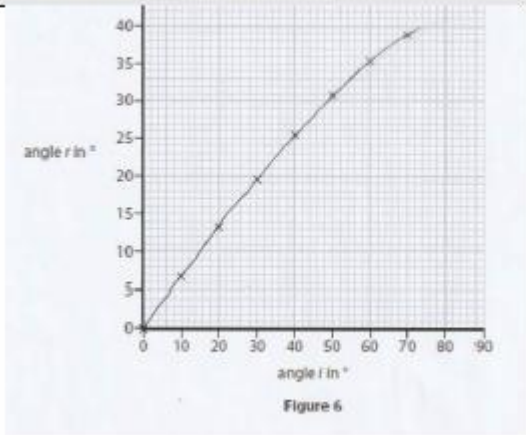
Q4.

Question number	Answer	Additional guidance	Marks
	An explanation that combines identification via a judgement (1 mark) to reach a conclusion via justification (1 mark):		(2)
	• anomalous result / it is at (angle of incidence) 40° (1)	the 4th cross the one in the middle. may be identified on the graph	
	• (because) it is further away from the curve of best fit than the other results (1)	does not follow same pattern as the others / not in line etc. explains the cause of the error e.g. misreading protractor suggests a plausible correct result such as 40(°),25(°) ignore points at i = 40(°) and 50(°) are the same	

Q5.

Question number	Answer	Additional guidance	Mark
	a description of method to include **four** from: draw around the block (1) shine a ray on the block (1) mark/draw/trace where the ray enters (1) mark/draw/trace where the ray exits (1) join point of entry to point of exit (1)	 point the ray box at the block can credit from drawing with glass block and ray box/light box labelled mark/draw/trace incident ray mark/draw/trace emerging ray allow 'draw light rays' for 1 mark if mp3 and mp4 not scored	**(4)** **AO1.2**

Q6.

Question number	Answer	Additional guidance	Mark
(i)	curve through origin, through all points – by eye (1)		(1) AO1

Question number	Answer	Additional guidance	Mark
(ii)	(***r*** =) 42(°) ± 2(°) (1)	ECF their graph	(1) AO3

Question number	Answer	Additional guidance	Mark
(iii)	Description to include two from: ***r*** increases as ***i*** increases (1) (but) not in proportion (1) increase in ***r*** becomes less (for same increase in ***i***) (1)	 ***r*** increases as ***i*** increases (but) not in even steps/not straight line/non-linear/gradient changes ***r*** always less than ***i***	(2) AO3